Computer Lab Assignments

In problems 49–52, use a CAS to find the sum of the given complex series. In *Mathematica* use the command **Sum** with **Infinity** as the upper limit. Use the command **N** to write your answer in the form $a + bi$.

49. $\displaystyle\sum_{k=1}^{\infty} \frac{i}{k(1+i)^k}$

50. $\displaystyle\sum_{k=1}^{\infty} \frac{i^k}{k^3}$

51. $\displaystyle\sum_{k=1}^{\infty} \frac{(k+1)i^k}{k!}$

52. $\displaystyle\sum_{k=1}^{\infty} \frac{k^2(1-i)^k}{2^k}$

6.2 Taylor Series

INTRODUCTION The correspondence between a complex number z within the circle of convergence and the number to which the series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ converges is single-valued. In this sense, a power series *defines* or *represents* a function f; for a specified z within the circle of convergence, the number L to which the power series converges is defined to be the value of f at z, that is, $f(z) = L$. In this section we present some important facts about the nature of this function f.

In the preceding section we saw that every power series has a radius of convergence R. Throughout the discussion in this section we will assume that a power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ has either a *positive* or an *infinite* radius R of convergence.

Differentiation and Integration of Power Series

The three theorems that follow indicate a function f that is defined by a power series is continuous, differentiable, and integrable within its circle of convergence.

THEOREM 6.2.1 Continuity

A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ represents a continuous function f within its circle of convergence $|z - z_0| = R$.

THEOREM 6.2.2 Term-by-Term Differentiation

A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ can be differentiated term by term within its circle of convergence $|z - z_0| = R$.

Differentiating a power series term-by-term gives,

$$\frac{d}{dz}\sum_{k=0}^{\infty} a_k(z - z_0)^k = \sum_{k=0}^{\infty} a_k \frac{d}{dz}(z - z_0)^k = \sum_{k=1}^{\infty} a_k k(z - z_0)^{k-1}.$$

Note that the summation index in the last series starts with $k = 1$ because the term corresponding to $k = 0$ is zero. It is readily proved by the ratio test that the original series and the differentiated series,

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k \qquad \text{and} \qquad \sum_{k=1}^{\infty} a_k k(z - z_0)^{k-1}$$

have the same circle of convergence $|z - z_0| = R$. Since the derivative of a power series is another power series, the first series $\sum_{k=1}^{\infty} a_k(z - z_0)^k$ can be differentiated as many times as we wish. In other words, it follows as a corollary to Theorem 6.2.2 that *a power series defines an infinitely differentiable function* within its circle of convergence and each differentiated series has the same radius of convergence R as the original power series.

Important ▶

THEOREM 6.2.3 Term-by-Term Integration

A power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ can be integrated term-by-term within its circle of convergence $|z - z_0| = R$, for every contour C lying entirely within the circle of convergence.

The theorem states that

$$\int_C \sum_{k=0}^{\infty} a_k(z - z_0)^k \, dz = \sum_{k=0}^{\infty} a_k \int_C (z - z_0)^k \, dz$$

whenever C lies in the interior of $|z - z_0| = R$. Indefinite integration can also be carried out term by term:

$$\int \sum_{k=0}^{\infty} a_k(z - z_0)^k dz = \sum_{k=0}^{\infty} a_k \int (z - z_0)^k dz$$
$$= \sum_{k=0}^{\infty} \frac{a_k}{k+1}(z - z_0)^{k+1} + \text{constant}.$$

The ratio test given in Theorem 6.1.4 can be used to be prove that both

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k \qquad \text{and} \qquad \sum_{k=0}^{\infty} \frac{a_k}{k+1}(z - z_0)^{k+1}$$

have the same circle of convergence $|z - z_0| = R$.

A review of real Taylor series can be found in Section 6.2 of the student study guide.

▶ **Taylor Series** Suppose a power series represents a function f within $|z - z_0| = R$, that is,

$$f(z) = \sum_{k=0}^{\infty} a_k(z - z_0)^k = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots . \tag{1}$$

It follows from Theorem 6.2.2 that the derivatives of f are the series

$$f'(z) = \sum_{k=1}^{\infty} a_k k(z - z_0)^{k-1} = a_1 + 2a_2(z - z_0) + 3a_3(z - z_0)^2 + \cdots , \tag{2}$$

$$f''(z) = \sum_{k=2}^{\infty} a_k k(k-1)(z - z_0)^{k-2} = 2 \cdot 1a_2 + 3 \cdot 2a_3(z - z_0) + \cdots , \tag{3}$$

$$f'''(z) = \sum_{k=3}^{\infty} a_k k(k-1)(k-2)(z - z_0)^{k-3} = 3 \cdot 2 \cdot 1a_3 + \cdots , \tag{4}$$

Important

▶ and so on. Since the power series (1) represents a differentiable function f within its circle of convergence $|z - z_0| = R$, where R is either a positive number or infinity, we conclude that a *power series represents an analytic function* within its circle of convergence.

There is a relationship between the coefficients a_k in (1) and the derivatives of f. Evaluating (1), (2), (3), and (4) at $z = z_0$ gives

$$f(z_0) = a_0, \quad f'(z_0) = 1!a_1, \quad f''(z_0) = 2!a_2, \quad \text{and} \quad f'''(z_0) = 3!a_3,$$

respectively. In general, $f^{(n)}(z_0) = n!\,a_n$, or

$$a_n = \frac{f^{(n)}(z_0)}{n!}, \quad n \geq 0. \tag{5}$$

When $n = 0$ in (5), we interpret the zero-order derivative as $f(z_0)$ and $0! = 1$ so that the formula gives $a_0 = f(z_0)$. Substituting (5) into (1) yields

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k. \tag{6}$$

This series is called the **Taylor series** for f centered at z_0. A Taylor series with center $z_0 = 0$,

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k, \tag{7}$$

is referred to as a **Maclaurin series**.

We have just seen that a power series with a nonzero radius R of convergence represents an analytic function. On the other hand we ask:

Question

If we are given a function f that is analytic in some domain D, can we represent it by a power series of the form (6) *or* (7)?

Since a power series converges in a circular domain, and a domain D is generally not circular, the question comes down to: Can we expand f in one or more power series that are *valid*—that is, a power series that converges at z and the number to which the series converges is $f(z)$—in circular domains that are all contained in D? The question is answered in the affirmative by the next theorem.

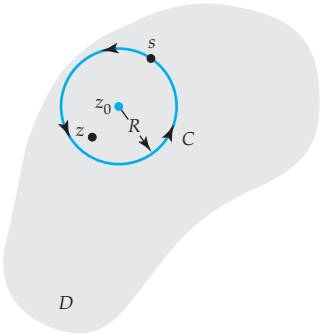


FIGURE 6.2.1 Contour for the proof of Theorem 6.2.4

THEOREM 6.2.4 Taylor's Theorem

Let f be analytic within a domain D and let z_0 be a point in D. Then f has the series representation

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k \tag{8}$$

valid for the largest circle C with center at z_0 and radius R that lies entirely within D.

Proof Let z be a fixed point within the circle C and let s denote the variable of integration. The circle C is then described by $|s - z_0| = R$. See FIGURE 6.2.1. To begin, we use the Cauchy integral formula to obtain the value of f at z:

$$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(s)}{s - z}ds = \frac{1}{2\pi i}\oint_C \frac{f(s)}{(s - z_0) - (z - z_0)}ds$$

$$= \frac{1}{2\pi i}\oint_C \frac{f(s)}{(s - z_0)}\left\{\frac{1}{1 - \dfrac{z - z_0}{s - z_0}}\right\}ds. \tag{9}$$

By replacing z by $(z - z_0)/(s - z_0)$ in (8) of Section 6.1, we have

$$\frac{1}{1 - \dfrac{z - z_0}{s - z_0}} = 1 + \frac{z - z_0}{s - z_0} + \left(\frac{z - z_0}{s - z_0}\right)^2 + \cdots + \left(\frac{z - z_0}{s - z_0}\right)^{n-1} + \frac{(z - z_0)^n}{(s - z)(s - z_0)^{n-1}},$$

and so (9) becomes

$$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(s)}{s-z_0}ds + \frac{z-z_0}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^2}ds$$
$$+ \frac{(z-z_0)^2}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^3}ds + \cdots + \frac{(z-z_0)^{n-1}}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^n}ds \qquad (10)$$
$$+ \frac{(z-z_0)^n}{2\pi i}\oint_C \frac{f(s)}{(s-z)(s-z_0)^n}ds.$$

Utilizing Cauchy's integral formula for derivatives, (6) of Section 5.5, we can rewrite (10) as

$$f(z) = f(z_0) + \frac{f'(z_0)}{1!}(z-z_0) + \frac{f''(z_0)}{2!}(z-z_0)^2 + \cdots + \frac{f^{(n-1)}(z_0)}{(n-1)!}(z-z_0)^{n-1} + R_n(z), \qquad (11)$$

where $$R_n(z) = \frac{(z-z_0)^n}{2\pi i}\oint_C \frac{f(s)}{(s-z)(s-z_0)^n}\,ds.$$

Equation (11) is called Taylor's formula with remainder R_n. We now wish to show that $R_n(z) \to 0$ as $n \to \infty$. This can be accomplished by showing that $|\,R_n(z)\,| \to 0$ as $n \to \infty$. Since f is analytic in D, by Theorem 5.5.8 we know that $|f(z)|$ has a maximum value M on the contour C. In addition, since z is inside C, $|z-z_0| < R$ and, consequently,

$$|s-z| = |s-z_0-(z-z_0)| \geq |s-z_0| - |z-z_0| = R-d,$$

where $d = |z-z_0|$ is the distance from z to z_0. The *ML*-inequality then gives

$$|\,R_n(z)\,| = \left|\frac{(z-z_0)^n}{2\pi i}\oint_C \frac{f(s)}{(s-z)(s-z_0)^n}\,ds\right| \leq \frac{d^n}{2\pi}\cdot\frac{M}{(R-d)R^n}\cdot 2\pi R = \frac{MR}{R-d}\left(\frac{d}{R}\right)^n.$$

Because $d < R$, $(d/R)^n \to 0$ as $n \to \infty$ we conclude that $|\,R_n(z)\,| \to 0$ as $n \to \infty$. It follows that the infinite series

$$f(z_0) + \frac{f'(z_0)}{1!}(z-z_0) + \frac{f''(z_0)}{2!}(z-z_0)^2 + \cdots$$

converges to $f(z)$. In other words, the result in (8) is valid for any point z interior to C. ☰

Note ▶ We can find the radius of convergence of a Taylor series in exactly the same manner illustrated in Examples 5–7 of the preceding section. However, we can simplify matters even further by noting that *the radius of convergence R is the distance from the center z_0 of the series to the nearest* ***isolated singularity*** *of f*. We shall elaborate more on this concept in the next section, but an isolated singularity is a point at which f fails to be analytic but is, nonetheless, analytic at all other points throughout *some* neighborhood of the point. For example, $z = 5i$ is an isolated singularity of $f(z) = 1/(z-5i)$. If the function f is entire, then the radius of convergence of a Taylor series centered at any point z_0 is necessarily $R = \infty$. Using (8) and the last fact, we can say that the following Maclaurin series representations are valid for all z, that is, for $|z| < \infty$.

Some Important Maclaurin Series

$$e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots = \sum_{k=0}^{\infty} \frac{z^k}{k!} \tag{12}$$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} \tag{13}$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} \tag{14}$$

EXAMPLE 1 Radius of Convergence

Suppose the function $f(z) = \dfrac{3-i}{1-i+z}$ is expanded in a Taylor series with center $z_0 = 4 - 2i$. What is its radius of convergence R?

Solution Observe that the function is analytic at every point except at $z = -1 + i$, which is an isolated singularity of f. The distance from $z = -1 + i$ to $z_0 = 4 - 2i$ is

$$|z - z_0| = \sqrt{(-1-4)^2 + (1-(-2))^2} = \sqrt{34}.$$

This last number is the radius of convergence R for the Taylor series centered at $4 - 2i$. ☰

If two power series with center z_0,

$$\sum_{k=0}^{\infty} a_k (z - z_0)^k \qquad \text{and} \qquad \sum_{k=0}^{\infty} b_k (z - z_0)^k$$

represent the same function f and have the same nonzero radius R of convergence, then

$$a_k = b_k = \frac{f^{(k)}(z_0)}{k!}, \quad k = 0, 1, 2, \ldots .$$

Note: Generally, the formula in (8) is used as a last resort. ▶

Stated in another way, the power series expansion of a function, with center z_0, is *unique*. On a practical level this means that a power series expansion of an analytic function f centered at z_0, irrespective of the method used to obtain it, is *the* Taylor series expansion of the function. For example, we can obtain (14) by simply differentiating (13) term by term. The Maclaurin series for e^{z^2} can be obtained by replacing the symbol z in (12) by z^2.

EXAMPLE 2 Maclaurin Series

Find the Maclaurin expansion of $f(z) = \dfrac{1}{(1-z)^2}$.

Solution We could, of course, begin by computing the coefficients using (8). However, recall from (6) of Section 6.1 that for $|z| < 1$,

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots . \tag{15}$$

If we differentiate both sides of the last result with respect to z, then

$$\frac{d}{dz}\frac{1}{1-z} = \frac{d}{dz}1 + \frac{d}{dz}z + \frac{d}{dz}z^2 + \frac{d}{dz}z^3 + \cdots$$

or
$$\frac{1}{(1-z)^2} = 0 + 1 + 2z + 3z^2 + \cdots = \sum_{k=1}^{\infty} k\, z^{k-1}. \tag{16}$$

Since we are using Theorem 6.2.2, the radius of convergence of the last power series is the same as the original series, $R = 1$. ☰

We can often build on results such as (16). For example, if we want the Maclaurin expansion of $f(z) = \dfrac{z^3}{(1-z)^2}$, we simply multiply (16) by z^3:

$$\frac{z^3}{(1-z)^2} = z^3 + 2z^4 + 3z^5 + \cdots = \sum_{k=1}^{\infty} k\, z^{k+2}.$$

The radius of convergence of the last series is still $R = 1$.

EXAMPLE 3 Taylor Series

Expand $f(z) = \dfrac{1}{1-z}$ in a Taylor series with center $z_0 = 2i$.

Solution In this solution we again use the geometric series (15). By adding and subtracting $2i$ in the denominator of $1/(1-z)$, we can write

$$\frac{1}{1-z} = \frac{1}{1-z+2i-2i} = \frac{1}{1-2i-(z-2i)} = \frac{1}{1-2i}\,\frac{1}{1-\dfrac{z-2i}{1-2i}}.$$

We now write $\dfrac{1}{1-\dfrac{z-2i}{1-2i}}$ as a power series by using (15) with the symbol z replaced by the expression $\dfrac{z-2i}{1-2i}$:

$$\frac{1}{1-z} = \frac{1}{1-2i}\left[1 + \frac{z-2i}{1-2i} + \left(\frac{z-2i}{1-2i}\right)^2 + \left(\frac{z-2i}{1-2i}\right)^3 + \cdots\right]$$

or
$$\frac{1}{1-z} = \frac{1}{1-2i} + \frac{1}{(1-2i)^2}(z-2i) + \frac{1}{(1-2i)^3}(z-2i)^2 + \cdots. \tag{17}$$

Because the distance from the center $z_0 = 2i$ to the nearest singularity $z = 1$ is $\sqrt{5}$, we conclude that the circle of convergence for (17) is $|z-2i| = \sqrt{5}$. This can be verified by the ratio test of the preceding section. ☰

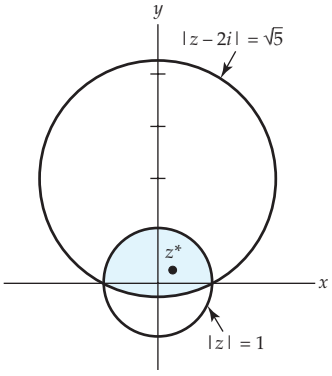


FIGURE 6.2.2 Series (15) and (17) both converge in the shaded region.

In (15) and (17) we represented the same function $f(z) = 1/(1-z)$ by two different power series. The first series (15) has center $z_0 = 0$ and radius of convergence $R = 1$. The second series (17) has center $z_0 = 2i$ and radius of convergence $R = \sqrt{5}$. The two different circles of convergence are illustrated in **FIGURE 6.2.2** The interior of the intersection of the two circles, shown in color, is the region where *both* series converge; in other words, at a specified point z^* in this region, both series converge to same value $f(z^*) = 1/(1-z^*)$. Outside the colored region at least one of the two series must diverge.

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) As a consequence of Theorem 5.5.3, we know that an analytic function f is infinitely differentiable. As a consequence of Theorem 6.2.4, we know that an analytic function f can always be expanded in a power series with a nonzero radius R of convergence. In real analysis, a function f can be infinitely differentiable, but it may be impossible to represent it by a power series. See Problem 55 in Exercises 6.2.

(*ii*) If you haven't already noticed, the results in (6), (7), (12), (13), and (14) are identical in form with their analogues in elementary calculus.

EXERCISES 6.2

Answers to selected odd-numbered problems begin on page ANS-16.

In Problems 1–12, use known results to expand the given function in a Maclaurin series. Give the radius of convergence R of each series.

1. $f(z) = \dfrac{z}{1+z}$

2. $f(z) = \dfrac{1}{4-2z}$

3. $f(z) = \dfrac{1}{(1+2z)^2}$

4. $f(z) = \dfrac{z}{(1-z)^3}$

5. $f(z) = e^{-2z}$

6. $f(z) = ze^{-z^2}$

7. $f(z) = \sinh z$

8. $f(z) = \cosh z$

9. $f(z) = \cos\dfrac{z}{2}$

10. $f(z) = \sin 3z$

11. $f(z) = \sin z^2$

12. $f(z) = \cos^2 z$ [*Hint*: Use a trigonometric identity.]

In Problems 13 and 14, use the Maclaurin series for e^z to expand the given function in a Taylor series centered at the indicated point z_0. [*Hint*: $z = z - z_0 + z_0$.]

13. $f(z) = e^z,\ z_0 = 3i$

14. $f(z) = (z-1)e^{-3z},\ z_0 = 1$

In Problems 15–22, expand the given function in a Taylor series centered at the indicated point z_0. Give the radius of convergence R of each series.

15. $f(z) = \dfrac{1}{z},\ z_0 = 1$

16. $f(z) = \dfrac{1}{z},\ z_0 = 1+i,$

17. $f(z) = \dfrac{1}{3-z},\ z_0 = 2i$

18. $f(z) = \dfrac{1}{1+z},\ z_0 = -i,$

19. $f(z) = \dfrac{z-1}{3-z},\ z_0 = 1$

20. $f(z) = \dfrac{1+z}{1-z},\ z_0 = i$

21. $f(z) = \cos z,\ z_0 = \pi/4$

22. $f(z) = \sin z,\ \ z_0 = \pi/2$

In Problems 23 and 24, use (7) to find the first three nonzero terms of the Maclaurin series of the given function.

23. $f(z) = \tan z$

24. $f(z) = e^{1/(1+z)}$

In Problems 25 and 26, use partial fractions as an aid in obtaining the Maclaurin series for the given function. Give the radius of convergence R of the series.

25. $f(z) = \dfrac{i}{(z-i)(z-2i)}$

26. $f(z) = \dfrac{z-7}{z^2-2z-3}$

In Problems 27 and 28, without actually expanding, determine the radius of convergence R of the Taylor series of the given function centered at the indicated point.

27. $f(z) = \dfrac{4+5z}{1+z^2},\ z_0 = 2+5i$ **28.** $f(z) = \cot z,\ z_0 = \pi i$

29. What is the radius of convergence R of the Maclaurin series in Problem 23?

30. What is the radius of convergence R of the Maclaurin series in Problem 24?

In Problems 31 and 32, expand the given function in Taylor series centered at each of the indicated points. Give the radius of convergence R of each series. Sketch the region within which both series converge.

31. $f(z) = \dfrac{1}{2+z},\ z_0 = -1,\ z_0 = i$ **32.** $f(z) = \dfrac{1}{z},\ z_0 = 1+i,\ z_0 = 3$

In Problems 33 and 34, use results obtained in this section to find the sum of the given power series.

33. $\displaystyle\sum_{k=0}^{\infty} 3^k z^k$ **34.** $\displaystyle\sum_{k=0}^{\infty} \frac{z^2}{k!}$

35. Find the Maclaurin series (14) by differentiating the Maclaurin series (13).

36. The **error function** $\operatorname{erf}(z)$ is defined by the integral $\operatorname{erf}(z) = \dfrac{2}{\sqrt{\pi}} \displaystyle\int_0^z e^{-t^2} dt$.

Find a Maclaurin series for $\operatorname{erf}(z)$ by integrating the Maclaurin series for e^{-t^2}.

In Problems 37 and 38, approximate the value of the given expression using the indicated number of terms of a Maclaurin series.

37. $e^{(1+i)/10}$, three terms **38.** $\sin\left(\dfrac{1+i}{10}\right)$, two terms

Focus on Concepts

39. Every function f has a domain of definition. Describe in words the domain of the function f defined by a power series center at z_0.

40. If $f(z) = \sum_{k=0}^{\infty} a_k z^k$ and $g(z) = \sum_{k=0}^{\infty} b_k z^k$ then the **Cauchy product** of f and g is given by

$$f(z)g(z) = \sum_{k=0}^{\infty} c_k z^k \quad \text{where} \quad c_k = \sum_{n=0}^{k} a_n b_{k-n}.$$

Write out the first five terms of the power series of $f(z)g(z)$.

41. Use Problem 40, (12) of this section, and (6) from Section 6.1 to find the first four nonzero terms of the Maclaurin series of $e^z/(1-z)$. What is the radius of convergence R of the series?

42. Use Problem 40, and (13) and (14) of this section to find the first four nonzero terms of the Maclaurin series of $\sin z \cos z$. Can you think of another way to obtain this series?

43. The function $f(z) = \sec z$ is analytic at $z = 0$ and hence possesses a Maclaurin series representation. We could, of course, use (7), but there are several alternative ways to obtain the coefficients of the series

$$\sec z = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots.$$

One way is to equate coefficients on both sides of the identity $1 = (\sec z)\cos z$ or

$$1 = (a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots)\left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots\right).$$

Find the first three nonzero terms of the Maclaurin series of f. What is the radius of convergence R of the series?

44. (**a**) Use the definition $f(z) = \sec z = 1/\cos z$ and long division to obtain the first three nonzero terms of the Maclaurin series in Problem 43.

(**b**) Use $f(z) = \csc z = 1/\sin z$ and long division to obtain the first three nonzero terms of an infinite series. Is this series a Maclaurin series?

45. Suppose that a complex function f is analytic in a domain D that contains $z_0 = 0$ and f satisfies $f'(z) = 4z + f^2(z)$. Suppose further that $f(0) = 1$.

(**a**) Compute $f'(0)$, $f''(0)$, $f'''(0)$, $f^{(4)}(0)$, and $f^{(5)}(0)$.

(**b**) Find the first six terms of the Maclaurin expansion of f.

46. Find an alternative way of finding the first three nonzero terms of the Maclaurin series for $f(z) = \tan z$ (see Problem 23):

(**a**) based on the identity $\tan z = \sin z \sec z$ and Problems 42 and 43

(**b**) based on Problem 44(a)

(**c**) based on Problem 45. [*Hint*: $f'(z) = \sec^2 z = 1 + \tan^2 z$.]

47. We saw in Problem 34 in Exercises 1.3 that de Moivre's formula can be used to obtain trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$. Discuss how these identities can be used to obtain Maclaurin series for $\sin^3 z$ and $\cos^3 z$. [*Hint*: You might want to simplify your answers to Problem 34. For example, $\cos^2\theta \sin\theta = (1 - \sin^2\theta)\sin\theta$.]

48. (**a**) Suppose that the principal value of the logarithm $\operatorname{Ln} z = \log_e |z| + i \operatorname{Arg}(z)$ is expanded in a Taylor series with center $z_0 = -1 + i$. Explain why $R = 1$ is the radius of the largest circle centered at $z_0 = -1 + i$ within which f is analytic.

(**b**) Show that within the circle $|z - (-1 + i)| = 1$ the Taylor series for f is

$$\operatorname{Ln} z = \frac{1}{2}\log_e 2 + \frac{3\pi}{4}i - \sum_{k=1}^{\infty} \frac{1}{k}\left(\frac{1+i}{2}\right)^k (z + 1 - i)^k.$$

(**c**) Show that the radius of convergence for the power series in part (b) is $R = \sqrt{2}$. Explain why this does not contradict the result in part (a).

49. (**a**) Consider the function $\operatorname{Ln}(1 + z)$. What is the radius of the largest circle centered at the origin within which f is analytic.

(**b**) Expand f in a Maclaurin series. What is the radius of convergence of this series?

(**c**) Use the result in part (b) to find a Maclaurin series for $\operatorname{Ln}(1 - z)$.

(**d**) Find a Maclaurin series for $\operatorname{Ln}\left(\dfrac{1+z}{1-z}\right)$.

50. In Theorem 3.2.3 we saw that L'Hôpital's rule carries over to complex analysis.

In Problem 33 in Exercises 3.2 you were guided through a proof of the following proposition by using the definition of the derivative:

If functions f and g are analytic at a point z_0 and $f(z_0) = 0$, $g(z_0) = 0$, but $g'(z_0) \neq 0$, then $\displaystyle\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$.

This time, prove the proposition by replacing $f(z)$ and $g(z)$ by their Taylor series centered at z_0.

Computer Lab Assignments

In problems 51–54, use a CAS to find the first three nonzero terms of the Taylor series of the given function centered at the indicated point z_0. In *Mathematica* use the command **Series**.

51. $f(z) = z^2 \tan z,\ z_0 = i$

52. $f(z) = \cos z \operatorname{Ln} z,\ z_0 = 1 + i$

53. $f(z) = \dfrac{\sin z}{i - z},\ z_0 = 1$

54. $f(z) = e^{\sinh(iz)} z,\ z_0 = \pi i$

Projects

55. (**a**) You will find the following real function in most older calculus texts:

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Do some reading in these calculus texts as an aid in showing that f is infinitely differentiable at every value of x. Show that f is not represented by its Maclaurin expansion at any value of $x \neq 0$.

(**b**) Investigate whether the complex analogue of the real function in part (a),

$$f(z) = \begin{cases} e^{-1/z^2}, & z \neq 0 \\ 0, & z = 0. \end{cases}$$

is infinitely differentiable at $z = 0$.

6.3 Laurent Series

INTRODUCTION If a complex function f fails to be analytic at a point $z = z_0$, then this point is said to be a **singularity** or **singular point** of the function. For example, the complex numbers $z = 2i$ and $z = -2i$ are singularities of the function $f(z) = z/(z^2 + 4)$ because f is discontinuous at each of these points. Recall from Section 4.1 that the principal value of the logarithm, Ln z, is analytic at all points except those points on the branch cut consisting of the nonpositive x-axis; that is, the branch point $z = 0$ as well as all negative real numbers are singular points of Ln z.

In this section we will be concerned with a new kind of "power series" expansion of f about an **isolated singularity** z_0. This new series will involve negative as well as nonnegative integer powers of $z - z_0$.

Isolated Singularities

Suppose that $z = z_0$ is a singularity of a complex function f. The point $z = z_0$ is said to be an **isolated singularity** of the function f if there exists *some* deleted neighborhood, or punctured open disk, $0 < |z - z_0| < R$ of z_0 throughout which f is analytic. For example, we have just seen that $z = 2i$ and $z = -2i$ are singularities of $f(z) = z/(z^2 + 4)$. Both $2i$ and $-2i$ are isolated singularities since f is analytic at every point in the neighborhood defined by $|z - 2i| < 1$, except at $z = 2i$, and at every point in the neighborhood defined by $|z - (-2i)| < 1$, except at $z = -2i$. In other words, f is analytic in the deleted neighborhoods $0 < |z - 2i| < 1$ and $0 < |z + 2i| < 1$. On the other hand, the branch point $z = 0$ is *not* an isolated singularity of Ln z since every neighborhood of $z = 0$ must contain points on the negative x-axis. We say that a singular point $z = z_0$ of a function f is **nonisolated** if *every* neighborhood of z_0 contains at least one singularity of f other than z_0. For example, the branch point $z = 0$ is a nonisolated singularity of Ln z.

A New Kind of Series

If $z = z_0$ is a singularity of a function f, then certainly f cannot be expanded in a power series with z_0 as its center. However, about

an isolated singularity $z = z_0$, it is possible to represent f by a series involving both negative and nonnegative integer powers of $z - z_0$; that is,

$$f(z) = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots . \quad (1)$$

As a *very* simple example of (1) let us consider the function $f(z) = 1/(z - 1)$. As can be seen, the point $z = 1$ is an isolated singularity of f and consequently the function cannot be expanded in a Taylor series centered at that point. Nevertheless, f can be expanded in a series of the form given in (1) that is valid for all z *near* 1:

$$f(z) = \cdots + \frac{0}{(z - 1)^2} + \frac{1}{z - 1} + 0 + 0 \cdot (z - 1) + 0 \cdot (z - 1)^2 + \cdots . \quad (2)$$

The series representation in (2) is valid for $0 < |z - 1| < \infty$.

Using summation notation, we can write (1) as the sum of two series

$$f(z) = \sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z - z_0)^k. \quad (3)$$

The two series on the right-hand side in (3) are given special names. The part with negative powers of $z - z_0$, that is,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \quad (4)$$

is called the **principal part** of the series (1) and will converge for $|1/(z - z_0)| < r^*$ or equivalently for $|z - z_0| > 1/r^* = r$. The part consisting of the nonnegative powers of $z - z_0$,

$$\sum_{k=0}^{\infty} a_k(z - z_0)^k, \quad (5)$$

is called the **analytic part** of the series (1) and will converge for $|z - z_0| < R$. Hence, the sum of (4) and (5) converges when z satisfies both $|z - z_0| > r$ *and* $|z - z_0| < R$, that is, when z is a point in an annular domain defined by $r < |z - z_0| < R$.

By summing over negative and nonnegative integers, (1) can be written compactly as

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k.$$

The principal part of the series (2) consists of exactly one nonzero term, whereas its analytic part consists of all zero terms. Our next example illustrates a series of the form (1) in which the principal part of the series also consists of a finite number of nonzero terms, but this time the analytic part consists of an infinite number of nonzero terms.

EXAMPLE 1 Series of the Form Given in (1)

The function $f(z) = \dfrac{\sin z}{z^4}$ is not analytic at the isolated singularity $z = 0$ and hence cannot be expanded in a Maclaurin series. However, $\sin z$ is an entire function, and from (13) of Section 6.2 we know that its Maclaurin series,

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \frac{z^9}{9!} - \cdots ,$$

converges for $|z| < \infty$. By dividing this power series by z^4 we obtain a series for f with negative and positive integer powers of z:

$$f(z) = \frac{\sin z}{z^4} = \overbrace{\frac{1}{z^3} - \frac{1}{3!\,z}}^{\text{principal part}} \overbrace{+ \frac{z}{5!} - \frac{z^3}{7!} + \frac{z^5}{9!} - \cdots}^{\text{analytic part}}. \tag{6}$$

The analytic part of the series in (6) converges for $|z| < \infty$. (Verify.) The principal part is valid for $|z| > 0$. Thus (6) converges for all z except at $z = 0$; that is, the series representation is valid for $0 < |z| < \infty$. ☰

A series representation of a function f that has the form given in (1), and (2) and (6) are such examples, is called a **Laurent series** or a **Laurent expansion** of f about z_0 on the annulus $r < |z - z_0| < R$.

THEOREM 6.3.1 Laurent's Theorem

Let f be analytic within the annular domain D defined by $r < |z - z_0| < R$. Then f has the series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k \tag{7}$$

valid for $r < |z - z_0| < R$. The coefficients a_k are given by

$$a_k = \frac{1}{2\pi i} \oint_C \frac{f(s)}{(s - z_0)^{k+1}}\, ds, \quad k = 0, \pm 1, \pm 2, \ldots, \tag{8}$$

where C is a simple closed curve that lies entirely within D and has z_0 in its interior. See FIGURE 6.3.1.

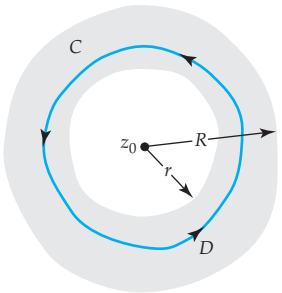


FIGURE 6.3.1 Contour for Theorem 6.3.1

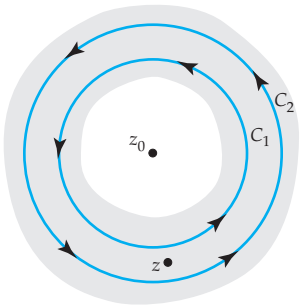


FIGURE 6.3.2 C_1 and C_2 are concentric circles.

Proof Let C_1 and C_2 be concentric circles with center z_0 and radii r_1 and R_2, where $r < r_1 < R_2 < R$. Let z be a fixed point in D that also satisfies the inequality $r_1 < |z - z_0| < R_2$. See FIGURE 6.3.2. By introducing a crosscut between C_2 and C_1 it follows from Cauchy's integral formula that

$$f(z) = \frac{1}{2\pi i} \oint_{C_2} \frac{f(s)}{s - z}\, ds - \frac{1}{2\pi i} \oint_{C_1} \frac{f(s)}{s - z}\, ds. \tag{9}$$

As in the proof of Theorem 6.2.4, we can write

$$\frac{1}{2\pi i} \oint_{C_2} \frac{f(s)}{s - z}\, ds = \sum_{k=0}^{\infty} a_k (z - z_0)^k, \tag{10}$$

where

$$a_k = \frac{1}{2\pi i} \oint_{C_2} \frac{f(s)}{(s - z_0)^{k+1}}\, ds, \quad k = 0, 1, 2, \ldots. \tag{11}$$

We then proceed in a manner similar to (9) of Section 6.2:

$$
\begin{aligned}
-\frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{s-z}\,ds &= \frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{(z-z_0)-(s-z_0)}\,ds \\
&= \frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{z-z_0}\left\{\frac{1}{1-\dfrac{s-z_0}{z-z_0}}\right\}ds \\
&= \frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{z-z_0}\left\{1+\frac{s-z_0}{z-z_0}+\left(\frac{s-z_0}{z-z_0}\right)^2+\cdots+\left(\frac{s-z_0}{z-z_0}\right)^{n-1}+\frac{(s-z_0)^n}{(z-s)(z-z_0)^{n-1}}\right\}ds \qquad (12)\\
&= \sum_{k=1}^{n}\frac{a_{-k}}{(z-z_0)^k}+R_n(z),
\end{aligned}
$$

where
$$a_{-k}=\frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{(s-z_0)^{-k+1}}\,ds,\quad k=1,\ 2,\ 3,\ \ldots, \qquad (13)$$

and
$$R_n(z)=\frac{1}{2\pi i(z-z_0)^n}\oint_{C_1}\frac{f(s)(s-z_0)^n}{z-s}\,ds.$$

Now let d denote the distance from z to z_0, that is, $|z-z_0|=d$, and let M denote the maximum value of $|f(z)|$ on the contour C_1. Using $|s-z_0|=r_1$ and the inequality (10) of Section 1.2,

$$|z-s|=|z-z_0-(s-z_0)|\geq |z-z_0|-|s-z_0|=d-r_1.$$

The *ML*-inequality then gives

$$
\begin{aligned}
|R_n(z)| &= \left|\frac{1}{2\pi i(z-z_0)^n}\oint_C\frac{f(s)(s-z_0)^n}{z-s}\,ds\right| \leq \frac{1}{2\pi d^n}\cdot\frac{Mr_1^n}{d-r_1}\cdot 2\pi r_1\\
&= \frac{Mr_1}{d-r_1}\left(\frac{r_1}{d}\right)^n.
\end{aligned}
$$

Because $r_1<d$, $(r_1/d)^n\to 0$ as $n\to\infty$, and so $|R_n(z)|\to 0$ as $n\to\infty$. Thus we have shown that

$$-\frac{1}{2\pi i}\oint_{C_1}\frac{f(s)}{s-z}\,ds=\sum_{k=1}^{\infty}\frac{a_{-k}}{(z-z_0)^k}, \qquad (14)$$

where the coefficients a_{-k} are given in (13). Combining (14) and (10), we see that (9) yields

$$f(z)=\sum_{k=1}^{\infty}\frac{a_{-k}}{(z-z_0)^k}+\sum_{k=0}^{\infty}a_k(z-z_0)^k. \qquad (15)$$

Finally, by summing over nonnegative and negative integers, (15) can be written as $f(z)=\sum_{k=-\infty}^{\infty}a_k(z-z_0)^k$. Moreover, (11) and (13) can be written as a single integral:

$$a_k=\oint_C\frac{f(z)}{(z-z_0)^{k+1}}\,dz,\quad k=0,\pm1,\pm2,\ldots,$$

where, in view of (5) of Section 5.3, we have replaced the contours C_1 and C_2 by any simple closed contour C in D with z_0 in its interior. ☰

In the case when $a_{-k}=0$ for $k=1, 2, 3, ...$, the principal part (4) is zero and the Laurent series (7) reduces to a Taylor series. Thus, a Laurent expansion can be considered as a generalization of a Taylor series.

The annular domain in Theorem 6.3.1 defined by $r < |z - z_0| < R$ need not have the "ring" shape illustrated in Figure 6.3.2. Here are some other possible annular domains:

$$(i)\ r = 0,\ R \text{ finite}, \quad (ii)\ r \neq 0,\ R = \infty, \text{ and } (iii)\ r = 0,\ R = \infty.$$

In the first case, the series converges in annular domain defined by an $0 < |z - z_0| < R$. This is the interior of the circle $|z - z_0| = R$ except the point z_0; in other words, the domain is a punctured open disk. In the second case, the annular domain is defined by $r < |z - z_0|$ and consists of all points exterior to the circle $|z - z_0| = r$. In the third case, the domain is defined by $0 < |z - z_0|$. This represents the entire complex plane except the point z_0. The Laurent series in (2) and (6) are valid on this last type of domain.

The coefficients defined by (8) are seldom used. See (*ii*) in the *Remarks* at the end of this section. ▶

The integral formula in (8) for the coefficients of a Laurent series are rarely used in actual practice. As a consequence, finding the Laurent series of a function in a specified annular domain is generally not an easy task. But this is not as disheartening as it might seem. In many instances we can obtain a desired Laurent series either by employing a known power series expansion of a function (as we did in Example 1) or by creative manipulation of geometric series (as we did in Example 2 of Section 6.2). The next example once again illustrates the use of geometric series.

EXAMPLE 2 Four Laurent Expansions

Expand $f(z) = \dfrac{1}{z(z-1)}$ in a Laurent series valid for the following annular domains.

(a) $0 < |z| < 1$ **(b)** $1 < |z|$ **(c)** $0 < |z - 1| < 1$ **(d)** $1 < |z - 1|$

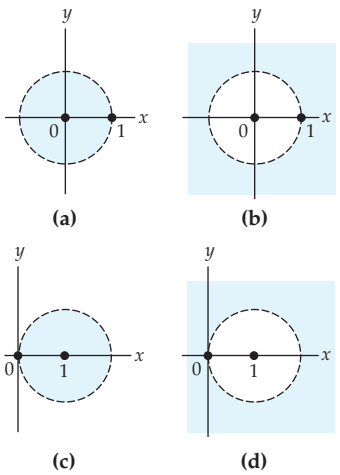


FIGURE 6.3.3 Annular domains for Example 2

Solution The four specified annular domains are shown in FIGURE 6.3.3. The black dots in each figure represent the two isolated singularities, $z = 0$ and $z = 1$, of f. In parts (a) and (b) we want to represent f in a series involving only negative and nonnegative integer powers of z, whereas in parts (c) and (d) we want to represent f in a series involving negative and nonnegative integer powers of $z - 1$.

(a) By writing

$$f(z) = -\frac{1}{z}\,\frac{1}{1-z},$$

we can use (6) of Section 6.1 to write $1/(1 - z)$ as a series:

$$f(z) = -\frac{1}{z}\left[1 + z + z^2 + z^3 + \cdots\right].$$

The infinite series in the brackets converges for $|z| < 1$, but after we multiply this expression by $1/z$, the resulting series

$$f(z) = -\frac{1}{z} - 1 - z - z^2 - z^3 - \cdots$$

converges for $0 < |z| < 1$.

(b) To obtain a series that converges for $1 < |z|$, we start by constructing a series that converges for $|1/z| < 1$. To this end we write the given function f as

$$f(z) = \frac{1}{z^2}\,\frac{1}{1-\frac{1}{z}}$$

and again use (6) of Section 6.1 with z replaced by $1/z$:

$$f(z) = \frac{1}{z^2}\left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right].$$

The series in the brackets converges for $|1/z| < 1$ or equivalently for $1 < |z|$. Thus the required Laurent series is

$$f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \cdots.$$

(**c**) This is basically the same problem as in part (a), except that we want all powers of $z - 1$. To that end, we add and subtract 1 in the denominator and use (7) of Section 6.1 with z replaced by $z - 1$:

$$\begin{aligned} f(z) &= \frac{1}{(1 - 1 + z)(z - 1)} \\ &= \frac{1}{z - 1}\,\frac{1}{1 + (z - 1)} \\ &= \frac{1}{z - 1}\left[1 - (z - 1) + (z - 1)^2 - (z - 1)^3 + \cdots\right] \\ &= \frac{1}{z - 1} - 1 + (z - 1) - (z - 1)^2 + \cdots. \end{aligned}$$

The requirement that $z \neq 1$ is equivalent to $0 < |z - 1|$, and the geometric series in brackets converges for $|z - 1| < 1$. Thus the last series converges for z satisfying $0 < |z - 1|$ *and* $|z - 1| < 1$, that is, for $0 < |z - 1| < 1$.

(**d**) Proceeding as in part (b), we write

$$\begin{aligned} f(z) &= \frac{1}{z - 1}\,\frac{1}{1 + (z - 1)} = \frac{1}{(z - 1)^2}\,\frac{1}{1 + \dfrac{1}{z - 1}} \\ &= \frac{1}{(z - 1)^2}\left[1 - \frac{1}{z - 1} + \frac{1}{(z - 1)^2} - \frac{1}{(z - 1)^3} + \cdots\right] \\ &= \frac{1}{(z - 1)^2} - \frac{1}{(z - 1)^3} + \frac{1}{(z - 1)^4} - \frac{1}{(z - 1)^5} + \cdots. \end{aligned}$$

Because the series within the brackets converges for $|\,1/(z - 1)\,| < 1$, the final series converges for $1 < |\,z - 1\,|$. ☰

EXAMPLE 3 Laurent Expansions

Expand $f(z) = \dfrac{1}{(z - 1)^2(z - 3)}$ in a Laurent series valid for (**a**) $0 < |\,z - 1\,| < 2$ and (**b**) $0 < |\,z - 3\,| < 2$.

Solution (**a**) As in parts (c) and (d) of Example 2, we want only powers of $z - 1$ and so we need to express $z - 3$ in terms of $z - 1$. This can be done by writing

$$f(z) = \frac{1}{(z - 1)^2(z - 3)} = \frac{1}{(z - 1)^2}\,\frac{1}{-2 + (z - 1)} = \frac{-1}{2(z - 1)^2}\,\frac{1}{1 - \dfrac{z - 1}{2}}$$

and then using (6) of Section 6.1 with the symbol z replaced by $(z-1)/2$,

$$f(z) = \frac{-1}{2(z-1)^2}\left[1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \cdots\right]$$

$$= -\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)} - \frac{1}{8} - \frac{1}{16}(z-1) - \cdots. \tag{16}$$

(**b**) To obtain powers of $z-3$, we write $z-1 = 2+(z-3)$ and

$$f(z) = \frac{1}{(z-1)^2(z-3)} = \frac{1}{z-3}\overbrace{[2+(z-3)]^{-2}}^{\text{We now factor 2 from this expression}}$$

$$= \frac{1}{4(z-3)}\left[1 + \frac{z-3}{2}\right]^{-2}.$$

At this point we can obtain a power series for $\left[1+\dfrac{z-3}{2}\right]^{-2}$ by using the binomial expansion,*

$$f(z) = \frac{1}{4(z-3)}\left[1 + \frac{(-2)}{1!}\left(\frac{z-3}{2}\right) + \frac{(-2)(-3)}{2!}\left(\frac{z-3}{2}\right)^2 + \frac{(-2)(-3)(-4)}{3!}\left(\frac{z-3}{2}\right)^3 + \cdots\right].$$

The binomial series in the brackets is valid for $|(z-3)/2| < 1$ or $|z-3| < 2$. Multiplying this series by $\dfrac{1}{4(z-3)}$ gives a Laurent series that is valid for $0 < |z-3| < 2$:

$$f(z) = \frac{1}{4(z-3)} - \frac{1}{4} + \frac{3}{16}(z-3) - \frac{1}{8}(z-3)^2 + \cdots. \quad \equiv$$

EXAMPLE 4 A Laurent Expansion

Expand $f(z) = \dfrac{8z+1}{z(1-z)}$ in a Laurent series valid for $0 < |z| < 1$.

Solution By partial fractions we can rewrite f as

$$f(z) = \frac{8z+1}{z(1-z)} = \frac{1}{z} + \frac{9}{1-z}.$$

Then by (6) of Section 6.1,

$$\frac{9}{1-z} = 9 + 9z + 9z^2 + \cdots.$$

The foregoing geometric series converges for $|z| < 1$, but after we add the term $1/z$ to it, the resulting Laurent series

$$f(z) = \frac{1}{z} + 9 + 9z + 9z^2 + \cdots$$

is valid for $0 < |z| < 1$. $\equiv$

In the preceding examples the point at the center of the annular domain of validity for each Laurent series was an isolated singularity of the function f. A re-examination of Theorem 6.3.1 shows that this need not be the case.

*For α real, the binomial series $(1+z)^\alpha = 1 + \alpha z + \dfrac{\alpha(\alpha-1)}{2!}z^2 + \dfrac{\alpha(\alpha-1)(\alpha-2)}{3!}z^3 + \cdots$ is valid for $|z| < 1$.

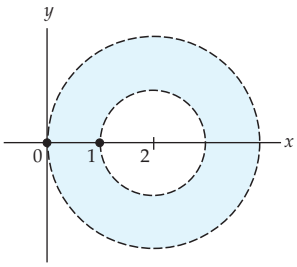


FIGURE 6.3.4 Annular domain for Example 5

EXAMPLE 5 A Laurent Expansion

Expand $f(z) = \dfrac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z-2| < 2$.

Solution The specified annular domain is shown in FIGURE 6.3.4. The center of this domain, $z = 2$, is the point of analyticity of the function f. Our goal now is to find two series involving integer powers of $z-2$, one converging for $1 < |z-2|$ and the other converging for $|z-2| < 2$. To accomplish this, we proceed as in the last example by decomposing f into partial fractions:

$$f(z) = -\frac{1}{z} + \frac{1}{z-1} = f_1(z) + f_2(z). \tag{17}$$

Now,
$$f_1(z) = -\frac{1}{z} = -\frac{1}{2+z-2}$$

$$= -\frac{1}{2}\,\frac{1}{1+\dfrac{z-2}{2}}$$

$$= -\frac{1}{2}\left[1 - \frac{z-2}{2} + \frac{(z-2)^2}{2^2} - \frac{(z-2)^3}{2^3} + \cdots\right]$$

$$= -\frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \cdots.$$

This series converges for $|(z-2)/2| < 1$ or $|z-2| < 2$. Furthermore,

$$f_2(z) = \frac{1}{z-1} = \frac{1}{1+z-2} = \frac{1}{z-2}\,\frac{1}{1+\dfrac{1}{z-2}}$$

$$= \frac{1}{z-2}\left[1 - \frac{1}{z-2} + \frac{1}{(z-2)^2} - \frac{1}{(z-2)^3} + \cdots\right]$$

$$= \frac{1}{z-2} - \frac{1}{(z-2)^2} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^4} + \cdots$$

converges for $|1/(z-2)| < 1$ or $1 < |z-2|$. Substituting these two results in (17) then gives

$$f(z) = \cdots - \frac{1}{(z-2)^4} + \frac{1}{(z-2)^3} - \frac{1}{(z-2)^2} + \frac{1}{z-2} - \frac{1}{2} + \frac{z-2}{2^2} - \frac{(z-2)^2}{2^3} + \frac{(z-2)^3}{2^4} - \cdots.$$

This representation is valid for z satisfying $|z-2| < 2$ and $1 < |z-2|$; in other words, for $1 < |z-2| < 2$. ☰

EXAMPLE 6 A Laurent Expansion

Expand $f(z) = e^{3/z}$ in a Laurent series valid for $0 < |z| < \infty$.

Solution From (12) of Section 6.2 we know that for all finite z, that is, $|z| < \infty$,

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots. \tag{18}$$

We obtain the Laurent series for f by simply replacing z in (18) by $3/z$, $z \neq 0$,

$$e^{3/z} = 1 + \frac{3}{z} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots. \tag{19}$$

This series (19) is valid for $z \neq 0$, that is, for $0 < |z| < \infty$.

Remarks

(*i*) In conclusion, we point out a result that will be of special interest to us in Sections 6.5 and 6.6. Replacing the complex variable s with the usual symbol z, we see that when $k = -1$, formula (8) for the Laurent series coefficients yields $a_{-1} = \dfrac{1}{2\pi i}\oint_C f(z)\,dz$, or more importantly,

$$\oint_C f(z)\,dz = 2\pi i\, a_{-1}. \tag{20}$$

(*ii*) Regardless how a Laurent expansion of a function f is obtained in a specified annular domain it is *the* Laurent series; that is, the series we obtain is unique.

EXERCISES 6.3

Answers to selected odd-numbered problems begin on page ANS-17.

In Problems 1–6, expand the given function in a Laurent series valid for the given annular domain.

1. $f(z) = \dfrac{\cos z}{z}$, $0 < |z|$

2. $f(z) = \dfrac{z - \sin z}{z^5}$, $0 < |z|$

3. $f(z) = e^{-1/z^2}$, $0 < |z|$

4. $f(z) = \dfrac{1 - e^z}{z^2}$, $0 < |z|$

5. $f(z) = \dfrac{e^z}{z - 1}$, $0 < |z - 1|$

6. $f(z) = z\cos\dfrac{1}{z}$, $0 < |z|$

In Problems 7–12, expand $f(z) = \dfrac{1}{z(z-3)}$ in a Laurent series valid for the indicated annular domain.

7. $0 < |z| < 3$

8. $0 < |z - 3| < 3$

9. $|z| > 3$

10. $|z - 3| > 3$

11. $1 < |z - 4| < 4$

12. $1 < |z + 1| < 4$

In Problems 13–16, expand $f(z) = \dfrac{1}{(z-1)(z-2)}$ in a Laurent series valid for the given annular domain.

13. $1 < |z| < 2$

14. $|z| > 2$

15. $0 < |z - 1| < 1$

16. $0 < |z - 2| < 1$

In Problems 17–20, expand $f(z) = \dfrac{z}{(z+1)(z-2)}$ in a Laurent series valid for the given annular domain.

17. $0 < |z + 1| < 3$

18. $|z + 1| > 3$

19. $1 < |z| < 2$

20. $0 < |z - 2| < 3$

In Problems 21 and 22, expand $f(z) = \dfrac{1}{z(1-z)^2}$ in a Laurent series valid for the given annular domain.

21. $0 < |z| < 1$

22. $|z| > 1$

In Problems 23 and 24, expand $f(z) = \dfrac{1}{(z-2)(z-1)^3}$ in a Laurent series valid for the given annular domain.

23. $0 < |z-2| < 1$

24. $0 < |z-1| < 1$

In Problems 25 and 26, expand $f(z) = \dfrac{7z-3}{z(z-1)}$ in a Laurent series valid for the given annular domain.

25. $0 < |z| < 1$

26. $0 < |z-1| < 1$ $\left[\textit{Hint: } \dfrac{7z-3}{z} = \dfrac{7(z-1)+4}{1+(z-1)}.\right]$

In Problems 27 and 28, expand $f(z) = \dfrac{z^2-2z+2}{z-2}$ in a Laurent series valid for the given annular domain.

27. $1 < |z-1|$

28. $0 < |z-2|$

In Problems 29 and 30, use $\cos z = 1 - \dfrac{z^2}{2!} + \dfrac{z^4}{4!} - \cdots$, $\sin z = z - \dfrac{z^3}{3!} + \dfrac{z^5}{5!} - \cdots$, and long division to find the first three nonzero terms of a Laurent series of the given function f valid for $0 < |z| < \pi$.

29. $f(z) = \csc z$

30. $f(z) = \cot z$

Focus on Concepts

31. The function $f(z) = \dfrac{1}{(z+2)(z-4i)}$ possesses a Laurent series $f(z) = \displaystyle\sum_{k=-\infty}^{\infty} a_k(z+2)^k$ valid in the annulus $r < |z+2| < R$. What are r and R?

32. Consider the function $f(z) = \dfrac{e^{-2z}}{(z+1)^2}$. Use (7) to find the principal part of the Laurent series expansion of f about $z_0 = -1$ that is valid on the annulus $0 < |z+1| < \infty$.

33. Consider the function $f(z) = \dfrac{1}{(z-5)^3}$. What is the Laurent series expansion of f about $z_0 = 5$ that is valid on the annulus $0 < |z-5| < \infty$?

34. Consider the function $f(z) = e^{(\alpha/2)(z-1/z)}$ where α is a real constant.

(**a**) Use (8) to show that the Laurent series expansion of f about $z_0 = 0$ is:

$$e^{(\alpha/2)(z-1/z)} = \sum_{k=-\infty}^{\infty} J_k(\alpha) z^k$$

where

$$J_k(\alpha) = \frac{1}{2\pi i} \oint_C \frac{e^{(\alpha/2)(s-1/s)}}{s^{k+1}}\, ds$$

and C is the unit circle.

(**b**) Use the parametrization $s(\theta) = e^{i\theta}$, $-\pi \le \theta \le \pi$ of C to show that

$$J_k(\alpha) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos\left(\alpha \sin(\theta) - k\theta\right) d\theta + \frac{i}{2\pi} \int_{-\pi}^{\pi} \sin\left(\alpha \sin(\theta) - k\theta\right) d\theta.$$

(**c**) Using integral properties of even and odd functions simplify the result in part (b) to obtain:

$$J_k(\alpha) = \frac{1}{\pi} \int_0^{\pi} \cos\left(k\theta - \alpha \sin\theta\right) d\theta.$$

The functions $J_k(\alpha)$ are called **Bessel functions** of the first kind of order k and are solutions to certain differential equations that occur frequently in applied mathematics.

Computer Lab Assignments

In problems 35–38, use a CAS to find the first four nonzero terms of the Laurent series of the given function centered at the indicated point z_0. In *Mathematica* the command **Series** finds Laurent series as well as Taylor series.

35. $f(z) = \dfrac{1}{z^2(z-i)^3(z+1)},\ z_0 = 0$

36. $f(z) = \dfrac{1}{z^2(z-i)^3(z+1)},\ z_0 = i$

37. $f(z) = \dfrac{\sin^2 z}{z^4},\ z_0 = 0$

38. $f(z) = \dfrac{\cosh^3(z) - 1}{\sin^3(z^2)}\, z,\ z_0 = 0$

6.4 Zeros and Poles

INTRODUCTION Suppose $z = z_0$ is an isolated singularity of a complex function f, and that

$$f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k = \sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} + \sum_{k=0}^{\infty} a_k(z - z_0)^k \tag{1}$$

is the Laurent series representation of f valid for the punctured open disk $0 < |z - z_0| < R$. We saw in the preceding section that a Laurent series (1) consists of two parts. That part of the series in (1) with negative powers of $z - z_0$, namely,

$$\sum_{k=1}^{\infty} a_{-k}(z - z_0)^{-k} = \sum_{k=1}^{\infty} \frac{a_{-k}}{(z - z_0)^k} \tag{2}$$

is the principal part of the series. In the discussion that follows we will assign different names to the isolated singularity $z = z_0$ according to the number of terms in the principal part.

Classification of Isolated Singular Points

An isolated singular point $z = z_0$ of a complex function f is given a classification depending on whether the principal part (2) of its Laurent expansion (1) contains zero, a finite number, or an infinite number of terms.

(*i*) If the principal part is zero, that is, *all* the coefficients a_{-k} in (2) are zero, then $z = z_0$ is called a **removable singularity**.

(*ii*) If the principal part contains a finite number of nonzero terms, then $z = z_0$ is called a **pole**. If, in this case, the last nonzero coefficient in (2) is a_{-n}, $n \geq 1$, then we say that $z = z_0$ is a **pole of order *n***. If $z = z_0$ is pole of order 1, then the principal part (2) contains exactly one term with coefficient a_{-1}. A pole of order 1 is commonly called a **simple pole**.

(*iii*) If the principal part (2) contains an infinitely many nonzero terms, then $z = z_0$ is called an **essential singularity**.

TABLE 6.4.1 summarizes the form of a Laurent series for a function f when $z = z_0$ is one of the above types of isolated singularities. Of course, R in the table could be ∞.

$z = z_0$	**Laurent Series for $0 < \lvert z - z_0 \rvert < R$**
Removable singularity	$a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$
Pole of order n	$\dfrac{a_{-n}}{(z - z_0)^n} + \dfrac{a_{-(n-1)}}{(z - z_0)^{n-1}} + \cdots + \dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots$
Simple pole	$\dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$
Essential singularity	$\cdots + \dfrac{a_{-2}}{(z - z_0)^2} + \dfrac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$

TABLE 6.4.1 Forms of Laurent series

EXAMPLE 1 Removable Singularity

Proceeding as we did in Example 1 of Section 6.3 by dividing the Maclaurin series for $\sin z$ by z, we see from

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots \tag{3}$$

that all the coefficients in the principal part of the Laurent series are zero. Hence $z = 0$ is a removable singularity of the function $f(z) = (\sin z)/z$.

Important paragraph. Reread it several times. ▶

If a function f has a removable singularity at the point $z = z_0$, then we can always supply an appropriate definition for the value of $f(z_0)$ so that f becomes analytic at $z = z_0$. For instance, since the right-hand side of (3) is 1 when we set $z = 0$, it makes sense to *define* $f(0) = 1$. Hence the function $f(z) = (\sin z)/z$, as given in (3), is now defined and continuous at every complex number z. Indeed, f is also analytic at $z = 0$ because it is represented by the Taylor series $1 - z^2/3! + z^4/5! - \cdots$ centered at 0 (a Maclaurin series).

EXAMPLE 2 Poles and Essential Singularity

(**a**) Dividing the terms of $\sin z = z - \dfrac{z^3}{3!} + \dfrac{z^5}{5!} - \cdots$ by z^2 shows that

$$\frac{\sin z}{z^2} = \overbrace{\frac{1}{z}}^{\text{principal part}} - \frac{z}{3!} + \frac{z^3}{5!} - \cdots$$

for $0 < |z| < \infty$. From this series we see that $a_{-1} \neq 0$ and so $z = 0$ is a simple pole of the function $f(z) = (\sin z)/z^2$. In like manner, we see that $z = 0$ is a pole of order 3 of the function $f(z) = (\sin z)/z^4$ considered in Example 1 of Section 6.3.

(**b**) In Example 3 of Section 6.3 we showed that the Laurent expansion of $f(z) = 1/(z - 1)^2(z - 3)$ valid for $0 < |z - 1| < 2$ was

$$f(z) = \overbrace{-\frac{1}{2(z-1)^2} - \frac{1}{4(z-1)}}^{\text{principal part}} - \frac{1}{8} - \frac{z-1}{16} - \cdots.$$

Since $a_{-2} = -\frac{1}{2} \neq 0$, we conclude that $z = 1$ is a pole of order 2.

(c) In Example 6 of Section 6.3 we see from (19) that the principal part of the Laurent expansion of the function $f(z) = e^{3/z}$ valid for $0 < |z| < \infty$ contains an infinite number of nonzero terms. This shows that $z = 0$ is an essential singularity of f. ☰

Zeros Recall, a number z_0 is **zero** of a function f if $f(z_0) = 0$. We say that an analytic function f has a **zero of order *n*** at $z = z_0$ if

$$\overbrace{f(z_0) = 0, \quad f'(z_0) = 0, \quad f''(z_0) = 0, \quad \ldots, \quad f^{(n-1)}(z_0) = 0}^{z_0 \text{ is a zero of } f \text{ and of its first } n-1 \text{ derivatives}}, \quad \text{but} \quad f^{(n)}(z_0) \neq 0. \tag{4}$$

A zero of order n is also referred to as a **zero of multiplicity *n***. For example, for $f(z) = (z - 5)^3$ we see that $f(5) = 0$, $f'(5) = 0$, $f''(5) = 0$, but $f'''(5) = 6 \neq 0$. Thus f has a zero of order (or multiplicity) 3 at $z_0 = 5$. A zero of order 1 is called a **simple zero**.

The next theorem is a consequence of (4).

THEOREM 6.4.1 Zero of Order *n*

A function f that is analytic in some disk $|z - z_0| < R$ has a zero of order n at $z = z_0$ if and only if f can be written

$$f(z) = (z - z_0)^n \phi(z), \tag{5}$$

where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

Partial Proof We will establish the "only if" part of the theorem. Given that f is analytic at z_0, it can be expanded in a Taylor series that is centered at z_0 and is convergent for $|z - z_0| < R$. Since the coefficients in a Taylor series $f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$ are $a_k = f^{(k)}(z_0)/k!$, $k = 0, 1, 2, \ldots$, it follows from (4) that the first n terms series are zero, and so the expansion must have the form

$$\begin{aligned} f(z) &= a_n (z - z_0)^n + a_{n+1}(z - z_0)^{n+1} + a_{n+2}(z - z_0)^{n+2} + \cdots \\ &= (z - z_0)^n \left[a_n + a_{n+1}(z - z_0) + a_{n+2}(z - z_0)^2 + \cdots \right]. \end{aligned}$$

With the power-series identification

$$\phi(z) = a_n + a_{n+1}(z - z_0) + a_{n+2}(z - z_0)^2 + \cdots$$

we conclude that ϕ is an analytic function and that $\phi(z_0) = a_n \neq 0$ because $a_n = f^{(n)}(z_0)/n! \neq 0$ from (4). ☰

EXAMPLE 3 Order of a Zero

The analytic function $f(z) = z \sin z^2$ has a zero at $z = 0$. If we replace z by z^2 in (13) of Section 6.2, we obtain the Maclaurin expansion

$$\sin z^2 = z^2 - \frac{z^6}{3!} + \frac{z^{10}}{5!} - \cdots.$$

Then by factoring z^2 out of the foregoing series we can rewrite f as

$$f(z) = z \sin z^2 = z^3 \phi(z) \qquad \text{where} \qquad \phi(z) = 1 - \frac{z^4}{3!} + \frac{z^8}{5!} - \cdots \tag{6}$$

is analytic and $\phi(0) = 1$. When compared to (5), the result in (6) shows that $z = 0$ is a zero of order 3 of f.

Poles

We can characterize a pole of order n in a manner analogous to (5).

THEOREM 6.4.2 Pole of Order n

A function f analytic in a punctured disk $0 < |z - z_0| < R$ has a pole of order n at $z = z_0$ if and only if f can be written

$$f(z) = \frac{\phi(z)}{(z - z_0)^n}, \tag{7}$$

where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$.

Partial Proof As in the proof of (5), we will establish the "only if" part of the preceding sentence. Since f is assumed to have a pole of order n at z_0 it can be expanded in a Laurent series

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots, \tag{8}$$

valid in the disk $0 < |z - z_0| < R$. By factoring out $1/(z - z_0)^n$, (8) confirms that f can be written in the form $\phi(z)/(z - z_0)^n$. Here we identify

$$\phi(z) = a_{-n} + \cdots + a_{-2}(z - z_0)^{n-2} + a_{-1}(z - z_0)^{n-1} + a_0(z - z_0)^n + a_1(z - z_0)^{n+1} + \cdots, \tag{9}$$

as a power series valid for the open disk $|z - z_0| < R$. By assumption, $z = z_0$ is a pole of order n of f, and so we must have $a_{-n} \neq 0$. If we define $\phi(z_0) = a_{-n}$, then it follows from (9) that ϕ is analytic throughout the disk $|z - z_0| < R$.

Zeros Again

A zero $z = z_0$ of an analytic function f is *isolated* in the sense that there exists some neighborhood of z_0 for which $f(z) \neq 0$ at every point z in that neighborhood except at $z = z_0$. As a consequence, if z_0 is a zero of a nontrivial analytic function f, then the function $1/f(z)$ has an isolated singularity at the point $z = z_0$.

The following result enables us, in some circumstances, to determine the poles of a function by inspection.

THEOREM 6.4.3 Pole of Order n

If the functions g and h are analytic at $z = z_0$ and h has a zero of order n at $z = z_0$ and $g(z_0) \neq 0$, then the function $f(z) = g(z)/h(z)$ has a pole of order n at $z = z_0$.

Proof Because the function h has zero of order n, (5) gives $h(z) = (z - z_0)^n \phi(z)$, where ϕ is analytic at $z = z_0$ and $\phi(z_0) \neq 0$. Thus f can be written

$$f(z) = \frac{g(z)/\phi(z)}{(z - z_0)^n}. \tag{10}$$

Since g and ϕ are analytic at $z = z_0$ and $\phi(z_0) \neq 0$, it follows that the function g/ϕ is analytic at z_0. Moreover, $g(z_0) \neq 0$ implies $g(z_0)/\phi(z_0) \neq 0$. We conclude from Theorem 6.4.2 that the function f has a pole of order n at z_0.

When $n = 1$ in (10), we see that a zero of order 1, or a simple zero, in the denominator h of $f(z) = g(z)/h(z)$ corresponds to a simple pole of f.

EXAMPLE 4 Order of Poles

(**a**) Inspection of the rational function

$$f(z) = \frac{2z + 5}{(z - 1)(z + 5)(z - 2)^4}$$

shows that the denominator has zeros of order 1 at $z = 1$ and $z = -5$, and a zero of order 4 at $z = 2$. Since the numerator is not zero at any of these points, it follows from Theorem 6.4.3 and (10) that f has simple poles at $z = 1$ and $z = -5$, and a pole of order 4 at $z = 2$.

(**b**) In Example 3 we saw that $z = 0$ is a zero of order 3 of $z \sin z^2$. From Theorem 6.4.3 and (10) we conclude that the reciprocal function $f(z) = 1/(z \sin z^2)$ has a pole of order 3 at $z = 0$. ☰

Remarks

(*i*) From the preceding discussion, it should be intuitively clear that if a function f has a pole at $z = z_0$, then $|f(z)| \to \infty$ as $z \to z_0$ from any direction. From (*i*) of the *Remarks* following Section 3.1 we can write $\lim_{z \to z_0} f(z) = \infty$.

(*ii*) If you peruse other texts on complex variables, and you are encouraged to do this, you may encounter the term *meromorphic*. A function f is **meromorphic** if it is analytic throughout a domain D, except possibly for poles in D. It can be proved that a meromorphic function can have at most a finite number of poles in D. For example, the rational function $f(z) = 1/(z^2 + 1)$ is meromorphic in the complex plane.

EXERCISES 6.4

Answers to selected odd-numbered problems begin on page ANS-18.

In Problems 1–4, show that $z = 0$ is a removable singularity of the given function. Supply a definition of $f(0)$ so that f is analytic at $z = 0$.

1. $f(z) = \dfrac{e^{2z} - 1}{z}$

2. $f(z) = \dfrac{z^3 - 4z^2}{1 - e^{z^2/2}}$

3. $f(z) = \dfrac{\sin 4z - 4z}{z^2}$

4. $f(z) = \dfrac{1 - \frac{1}{2}z^{10} - \cos z^5}{\sin z^2}$

In Problems 5–10, determine the zeros and their order for the given function.

5. $f(z) = (z + 2 - i)^2$

6. $f(z) = z^4 - 16$

7. $f(z) = z^4 + z^2$

8. $f(z) = \sin^2 z$

9. $f(z) = e^{2z} - e^z$

10. $f(z) = ze^z - z$

In Problems 11–14, the indicated number is a zero of the given function. Use a Maclaurin or Taylor series to determine the order of the zero.

11. $f(z) = z(1 - \cos^2 z);\ z = 0$

12. $f(z) = z - \sin z;\ z = 0$

13. $f(z) = 1 - e^{z-1};\ z = 1$

14. $f(z) = 1 - \pi i + z + e^z;\ z = \pi i$

In Problems 15–26, determine the order of the poles for the given function.

15. $f(z) = \dfrac{3z - 1}{z^2 + 2z + 5}$

16. $f(z) = 5 - \dfrac{6}{z^2}$

17. $f(z) = \dfrac{1 + 4i}{(z + 2)(z + i)^4}$

18. $f(z) = \dfrac{z - 1}{(z + 1)(z^3 + 1)}$

19. $f(z) = \tan z$

20. $f(z) = \dfrac{\cot \pi z}{z^2}$

21. $f(z) = \dfrac{1 - \cosh z}{z^4}$

22. $f(z) = \dfrac{e^z}{z^2}$

23. $f(z) = \dfrac{1}{1 + e^z}$

24. $f(z) = \dfrac{e^z - 1}{z^2}$

25. $f(z) = \dfrac{\sin z}{z^2 - z}$

26. $f(z) = \dfrac{\cos z - \cos 2z}{z^6}$

In Problems 27 and 28, show that the indicated number is an essential singularity of the given function.

27. $f(z) = z^3 \sin\left(\dfrac{1}{z}\right);\ z = 0$

28. $f(z) = (z - 1)\cos\left(\dfrac{1}{z + 2}\right);\ z = -2$

29. Determine whether $z = 0$ is an essential singularity of $f(z) = e^{z+1/z}$.

30. Determine whether $z = 0$ is an isolated or nonisolated singularity of $f(z) = \tan(1/z)$.

Focus on Concepts

31. In part (b) of Example 2 in Section 6.3, we showed that the Laurent series representation of $f(z) = \dfrac{1}{z(z - 1)}$ valid for $|z| > 1$ is

$$f(z) = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \cdots.$$

The point $z = 0$ is an isolated singularity of f, and the Laurent series contains an infinite number of terms involving negative integer powers of z. Discuss: Does this mean that $z = 0$ is an essential singularity of f? Defend your answer with sound mathematics.

32. Suppose f and g are analytic functions and f has a zero of order m and g has zero of order n at $z = z_0$. Discuss: What is the order of the zero of fg at z_0? of $f + g$ at z_0?

33. An interesting theorem, known as **Picard's theorem,** states that in any arbitrarily small neighborhood of an isolated essential singularity z_0, an analytic function f assumes *every* finite complex value, with *one* exception, an infinite number of times. Since $z = 0$ is an isolated essential singularity of $f(z) = e^{1/z}$, find an infinite number of z in any neighborhood of $z = 0$ for which $f(z) = i$. What is the one exception? That is, what is the one value that $f(z) = e^{1/z}$ does not take on?

34. Suppose $|f(z)|$ is bounded in a deleted neighborhood of an isolated singularity z_0. Classify z_0 as one of the three kinds of isolated singularities listed on page 289. Justify your answer with sound mathematics.

35. Suppose the analytic function $f(z)$ has a zero of order n at $z = z_0$. Prove that the function $[f(z)]^m$, m a positive integer, has a zero of order mn at $z = z_0$.

36. In this problem you are guided through the start of the proof of the proposition:

The only isolated singularities of a rational function f are poles or removable singularities.

Proof We begin with the hypothesis that f is a rational function, that is, $f(z) = p(z)/q(z)$, where p and q are polynomials. We know that f is analytic for all z except at the zeros of q. Suppose z_0 is a zero of q but not of p. Then Theorem 6.4.1 tell us that there exists a positive integer n such that $q(z) = (z - z_0)^n Q(z)$, where Q is a polynomial and $Q(z_0) \neq 0$. Now invoke Theorem 6.4.2. Consider one more case to finish the proof.

6.5 Residues and Residue Theorem

INTRODUCTION We saw in the last section that if a complex function f has an *isolated singularity* at a point z_0, then f has a Laurent series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k = \cdots + \frac{a_{-2}}{(z - z_0)^2} + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + \cdots,$$

which converges for all z *near* z_0. More precisely, the representation is valid in some deleted neighborhood of z_0 or punctured open disk $0 < |z - z_0| < R$. In this section our entire focus will be on the coefficient a_{-1} and its importance in the evaluation of contour integrals.

Residue The coefficient a_{-1} of $1/(z - z_0)$ in the preceding Laurent series is called the **residue** of the function f at the isolated singularity z_0. We shall use the notation

$$a_{-1} = \text{Res}(f(z),\ z_0)$$

to denote the residue of f at z_0. Recall, if the principal part of the Laurent series valid for $0 < |z - z_0| < R$ contains a finite number of terms with a_{-n} the last nonzero coefficient, then z_0 is a pole of order n; if the principal part of the series contains an infinite number of terms with nonzero coefficients, then z_0 is an essential singularity.

EXAMPLE 1 Residues

(**a**) In part (b) of Example 2 in Section 6.4 we saw that $z = 1$ is a pole of order two of the function $f(z) = \dfrac{1}{(z-1)^2(z-3)}$. From the Laurent series obtained in that example valid for the deleted neighborhood of $z = 1$ defined by $0 < |z - 1| < 2$,

$$f(z) = \frac{-1/2}{(z-1)^2} + \overbrace{\frac{-1/4}{z-1}}^{a_{-1}} - \frac{1}{8} - \frac{z-1}{16} - \cdots$$

we see that the coefficient of $1/(z - 1)$ is $a_{-1} = \text{Res}(f(z), 1) = -\frac{1}{4}$.

(**b**) In Example 6 of Section 6.3 we saw that $z = 0$ is an essential singularity of $f(z) = e^{3/z}$. Inspection of the Laurent series obtained in that example,

$$e^{3/z} = 1 + \overbrace{\frac{3}{z}}^{a_{-1}} + \frac{3^2}{2!z^2} + \frac{3^3}{3!z^3} + \cdots,$$

$0 < |z| < \infty$, shows that the coefficient of $1/z$ is $a_{-1} = \text{Res}(f(z),\ 0) = 3$. ☰

We will see why the coefficient a_{-1} is so important later on in this section. In the meantime we are going to examine ways of obtaining this complex number when z_0 is a *pole* of a function f without the necessity of expanding f in a Laurent series at z_0. We begin with the residue at a simple pole.